

FluxDash Software Documentation

Version 1.0.0 / Eddy Covariance Workflow, Teaching, and Research Support

Author and creator: Ife Familusi, Research Associate, Kentucky State University

Contact: ife.familusi@kysu.edu | Copyright © 2026 Ife Familusi. All rights reserved.

1. Purpose and scope

FluxDash is a Windows desktop application created to help students, early-career professionals, instructors, and researchers learn and apply eddy covariance workflows. The software brings data viewing, extraction, processing, pre-filtering, quality control, gap-filling, flux partitioning, interactive analysis, graph export, and workflow documentation into one guided interface.

FluxDash is designed as a teaching and research-support tool. It shows users what goes into each analytical stage without forcing them to manually coordinate several software environments at the beginning of their training. Users remain responsible for reviewing input data, instrument metadata, site context, QC flags, variable units, sign conventions, and model outputs before using results in reports, theses, publications, or management decisions.

2. Supported datasets

Input type	Typical files	FluxDash handling
LI-COR .ghg with embedded SmartFlux/EddyPro output	.ghg archive containing EddyPro output tables and biomet data	Extracts flux, meteorological, footprint, diagnostic, and biomet variables; applies user-selected pre-filtering before REddyProc.
Older raw .ghg files	.ghg archive containing high-frequency .data plus metadata and biomet files	Uses the built-in raw covariance fallback for teaching/exploratory processing; publication-grade results should be validated against EddyPro.
EddyPro/SmartFlux CSV outputs	EP-Summary or full output CSV files	Normalizes common NEE, H, LE, u*, radiation, Tair, VPD, RH, footprint, and QC variables.
FLUXNET/AmeriFlux products	FULLSET_HH, FULLSET_DD, FULLSET_MM, FULLSET_WW, FULLSET_YY, ERA5 products	Maps standardized products such as NEE_VUT_REF, GPP_NT_VUT_REF, RECO_NT_VUT_REF, LE_F_MDS, H_F_MDS, SW_IN_F_MDS, TA_F_MDS, VPD_F_MDS, and USTAR to FluxDash internal names.
Generic CSV exports	Timestamped CSV with flux and meteorological variables	Uses column mapping and available variables for visualization and analysis when required fields are present.

3. Installation and supporting environments

Install FluxDash using the Windows installer supplied through the official GitHub Releases page. For the recommended student workflow, install Python 3 and R before running more advanced processing and partitioning functions.

Component	Purpose	Notes
FluxDash Windows installer	Installs the desktop application.	Use the installer asset from the GitHub release, not the source-code archive automatically generated by GitHub.
Python 3	Supports the built-in .ghg extraction and	During installation, select Add Python to

	raw covariance fallback workflow.	PATH.
R	Required for REddyProc-based u* filtering, MDS gap filling, and partitioning.	FluxDash can help check/install the REddyProc package.
REddyProc R package	Performs gap filling and flux partitioning in the Partition tab.	Used for MDS gap filling and nighttime/daytime partitioning workflows.
EddyPro	Optional advanced processor.	Use only if external EddyPro reprocessing is specifically desired.

4. Recommended workflow

1. Open FluxDash and choose the data folder.
2. Use Scan folder to detect .ghg, EP-Summary, FLUXNET/AmeriFlux, or CSV datasets.
3. Review the detected date range and variables in the View tab.
4. Open the Process tab and select the processing engine.
5. Choose a pre-filtering/QC preset and review the variable-specific limits before processing.
6. Run processing and inspect the processed CSV, filter counts, and warnings.
7. Open the Partition tab and select the REddyProc method and u* option.
8. Run partitioning and verify gap-filling statistics before interpreting annual or cumulative budgets.
9. Use the Results tab to visualize NEE, GPP, Reco, EBC, ET, WUE, CUE, footprint, light response, Q10, seasonal patterns, and other derived analyses.
10. Export graphs, tables, partitioned files, and the workflow log.

5. Pre-filtering and quality-control system

FluxDash now includes a stronger pre-filtering and QC panel designed to teach users how raw or processed flux observations are screened before gap filling and partitioning. Rejected values are set to NA by default while timestamps are preserved, allowing REddyProc to operate on a regular half-hour grid.

Preset	Accepted QC flags	Suggested use
Strict	0 only	Conservative screening for high-confidence data and formal analyses when data coverage remains adequate.
Moderate	0 and 1	Recommended teaching default; keeps best and acceptable records while excluding poor-quality flags.
Permissive	0, 1, and 2	Exploratory visual inspection; users should document the choice before reporting results.
All permitted	All available flags	Raw-data exploration only; not recommended for final budgets.
Custom	User-defined	Allows site-specific or software-specific flag systems.

Variable group	QC/source column examples	How FluxDash applies the flag
CO ₂ / NEE	qc_co2_flux	Masks NEE/co2_flux only; it does not automatically remove H or LE.
Sensible heat	qc_H	Masks H only.
Latent heat	qc_LE	Masks LE only.
Water vapor	qc_h2o_flux	Masks h2o_flux only when present.
Momentum	qc_Tau	Masks Tau only when present.
Methane	qc_ch4_flux	Masks ch4_flux only when present.

Variable	Default lower bound	Default upper bound	Action
NEE / co2_flux	-50 umol m ⁻² s ⁻¹	+50 umol m ⁻² s ⁻¹	Set NEE to NA; preserve

			original value.
LE	-100 W m ⁻²	+1000 W m ⁻²	Set LE to NA; preserve original value.
H	-250 W m ⁻²	+800 W m ⁻²	Set H to NA; preserve original value.
Rn	-200 W m ⁻²	+1200 W m ⁻²	Set Rn to NA if outside plausible range.
Rg / SW_IN	0 W m ⁻²	+1400 W m ⁻²	Clamp small negative nighttime drift to 0 when selected; otherwise set invalid values to NA.
u*	0 m s ⁻¹	3 m s ⁻¹	Set invalid u* values to NA.
Tair	-50 C	+60 C	Set invalid air temperature to NA.
VPD	0 kPa	10 kPa	Clamp small negative values to 0 or set invalid values to NA.
RH	0%	100%	Cap RH above 100% to 100% when selected.

6. Instrument diagnostics and slow-driver cleaning

FluxDash can screen optical and signal diagnostics when they are available in the dataset. These checks are configurable because column names differ among instruments and output formats. Common examples include AGC, RSSI, signal strength, LI-7200 diagnostics, and LI-7500 signal metrics.

Rule	Default behavior	Important caution
AGC / optical diagnostic	Mask selected fast flux outputs when AGC exceeds 65%.	Use only when the dataset truly reports AGC in comparable units.
RSSI / signal strength	Mask selected fast flux outputs when RSSI or signal strength is below 80%.	Column definitions vary; review instrument documentation.
Negative nighttime radiation	Clamp $R_g < 0$ to 0 when selected.	Do not confuse Rg/SW_IN with net radiation Rn.
RH saturation	Cap $RH > 100\%$ at 100%.	Values far outside range may indicate sensor error and should be inspected.
VPD under saturation	Set VPD to 0 when RH is 100% if selected.	Check VPD units; FluxDash normalizes hPa/kPa/Pa where possible.
MAD driver flagging	Flag meteorological driver outliers using 4 x MAD by default.	Default is flag-only so true extreme events are not automatically removed.

7. Nighttime u* mask, gap filling, and partitioning

FluxDash separates pre-processing QC from REdDyProc partitioning. A user may optionally apply a pre-partitioning nighttime u* NEE-only mask, but the main REdDyProc workflow still handles u* threshold estimation, gap filling, and partitioning when the Partition tab is used.

Process	FluxDash behavior	Reporting reminder
Nighttime u* NEE-only mask	For records with $R_g < 10 \text{ W m}^{-2}$ and u* below the selected threshold, mask nighttime NEE only. H, LE, and daytime NEE are not removed using this rule.	Report the threshold, radiation criterion, and whether the rule was applied before partitioning.
Automatic u* estimation	Uses REdDyProc when possible. If automatic estimation is not finite, FluxDash falls back to the manual threshold and records this in the log.	Report the final threshold and any fallback.
MDS gap filling	Gap fills NEE and drivers through REdDyProc. FluxDash records missing-before and gap-filled counts.	Review the gap-filling completeness before using annual budgets.
Nighttime/Reichstein partitioning	Produces Reco and GPP based on nighttime temperature-response	State the method in manuscript or lab report.

	relationships.	
Daytime/Lasslop partitioning	Produces daytime light-response partitioning variables when the Lasslop method is selected.	If automatic u* fails, confirm the fallback manual threshold in the workflow log.

8. Full output preservation

FluxDash is designed to preserve traceability from original input through cleaned, gap-filled, and partitioned output. The final partitioned file attempts to retain original/source variables, processed/renamed variables, pre-filter flags, reject-reason columns, gap-filling flags, partitioning columns, derived variables, and footprint columns.

Output category	Examples
Original/source columns	Original instrument variables, biomet variables, albedo if available, footprint percentiles, diagnostic variables.
Preserved original values	NEE_orig_input, H_orig_input, LE_orig_input, Rn_orig_input, Rg_orig_input, VPD_orig_input.
Cleaned/processed values	co2_flux, H, LE, Rn, Rg, Tair, VPD, RH after selected cleaning rules.
Pre-filter/reject flags	NEE_prefilter_flag, H_prefilter_flag, LE_prefilter_flag, Rg_prefilter_flag, reject-reason columns.
Gap-filled values	NEE_uStar_f, NEE_U50_f, NEE_gapfilled_flag, driver-filled variables where available.
Partitioned values	GPP_uStar_f, Reco_uStar, GPP_DT_uStar, Reco_DT_uStar and uncertainty columns.
Derived analysis variables	ET, NEP, CUE, WUE, Bowen ratio, EBC, albedo, available energy, turbulent energy.

9. Results Explorer

The Results Explorer provides interactive analysis after processing and partitioning. Users can select presets, date ranges, grouping periods, summary statistics, unit conversions, outlier-handling options, and downloadable graphs/tables.

Analysis preset	Purpose
Cumulative NEE/GPP/Reco	Shows cumulative carbon exchange, source/sink status, and annual or seasonal budgets. NEE now prefers gap-filled REddyProc columns before falling back to original co2_flux.
Light response	Plots NEE or GPP against SW/Rg, PAR, or PPFD. Users can choose raw points, binned means, fitted curve, and robust-envelope fitting.
Temperature response / Q10	Relates Reco to air or soil temperature and reports fitted Q10-style parameters.
Energy balance closure	Plots or summarizes (H + LE) relative to available energy such as Rn - G, reports ratio, slope, intercept, R2, and n when available.
Gap-filling diagnostics	Summarizes original missing NEE, pre-filtered NEE, gap-filled NEE, and remaining data completeness.
Footprint explorer	Visualizes EddyPro/Kljun footprint percentile outputs and wind-direction source-area diagnostics.
Seasonal and diel patterns	Displays monthly, seasonal, and mean diel cycles for NEE, GPP, Reco, H, LE, ET, and related variables.

10. Recent v1.0.0 maintenance updates

- Results mapping now prefers gap-filled NEE columns such as NEE_U50_f or NEE_uStar_f for cumulative NEE and carbon-budget outputs.

- Light-response fitting now includes a robust-envelope fitting basis so GPP vs PPFD/PAR/SW better follows a rising-to-saturation pattern when the selected period supports it.
- Lasslop/daytime partitioning output handling was made safe when automatic u^* estimation fails and the workflow falls back to the manual threshold.
- VPD display now prefers normalized VPD columns to reduce Pa/hPa/kPa scale confusion.
- The workflow log records QC presets, accepted flags, diagnostic thresholds, MAD driver flagging, nighttime u^* NEE-only masks, source-column mappings, fit basis, and supporting references.

11. Light-response interpretation

A biologically reasonable GPP light-response curve often starts near zero at low light, increases rapidly at low to moderate radiation, and approaches saturation at high radiation. If the selected period is early-season, sparse, filtered heavily, or dominated by changing phenology, temperature, or VPD, the apparent relationship may look more linear and may not show a clear plateau. FluxDash therefore reports the fitted equation and parameters, but users should inspect binned means and data coverage before interpreting GPPmax or alpha.

Response	Expected direction	Recommended fit basis
GPP vs PPFD/PAR/SW	Positive GPP rises with light and may plateau at high radiation.	Robust envelope bins or binned means.
NEE vs PPFD/PAR/SW	NEE often becomes more negative as light increases under uptake conditions.	Lower-envelope bins or binned means.
Early-season or partial-year data	May not show clear saturation because canopy, VPD, and temperature also change.	Use seasonal subsets and inspect binned means.

12. Energy balance closure

Energy balance closure should use net radiation, not incoming shortwave radiation. The recommended x-axis is available energy and the y-axis is turbulent energy:

- Available energy = $R_n - G$ or $R_n - \text{mean}(\text{SHF plates})$
- Turbulent energy = $H + LE$

Energy balance closure can be reported as both a regression slope/intercept/ R^2 and as a ratio of sums: $\text{sum}(H + LE) / \text{sum}(R_n - G)$. Users should check whether G already represents the soil heat flux before subtracting both G and averaged soil heat-flux plates.

13. Workflow log and manuscript support

The exportable workflow log is intended to help users report exactly what was done. It includes dataset metadata, processing engine, date range, site settings, pre-filter/QC settings, filter counts, partitioning method, u^* settings, gap statistics, source-column mappings, results-view settings, and references supporting the selected workflow.

- State NEE sign convention: negative NEE indicates net ecosystem CO_2 uptake/sink.
- Report pre-filter limits, missing-value codes, QC flag acceptance, and whether rejected values were set to NA before gap filling.
- Report u^* method/value, gap-filling software, gap fraction, and partitioning method.
- For light-response curves, state whether NEE or GPP was plotted and whether SW, PAR, or PPFD was used.
- For footprint figures, state whether the plot used EddyPro/Kljun footprint percentile outputs or a full FFP recomputation.

14. Troubleshooting

Issue	Likely cause	Suggested response
No valid results produced from .ghg files	Files may contain only raw high-frequency data without embedded EddyPro output.	Use the built-in raw fallback or process through EddyPro; inspect one representative .ghg archive.
Automatic u* result not finite	Too few valid nighttime NEE records or too much filtering.	Use a defensible manual threshold and document the fallback in the log.
Cumulative NEE appears to have gaps	Results may be using original NEE rather than gap-filled NEE, or selected columns/date range are incomplete.	Use the latest build; verify source-column mapping and gap-filling diagnostics.
Light-response curve does not plateau	Partial season, insufficient high-light saturation, phenological change, or fit basis not appropriate.	Use robust-envelope bins, seasonal subsets, and inspect binned means before interpreting parameters.
Energy balance closure slope is low	Possible wrong radiation column, double-counted G/soil heat flux, sign convention issue, poor QC, or missing storage terms.	Use Rn rather than Rg/SW_IN, compare Rn-G vs Rn-SHFmean, and filter complete daytime records.
Lasslop output error after R completes	UI handling of null u* CI or missing mapping.	Use the latest maintenance build; check workflow log for fallback threshold and output columns.

15. References and methodological basis

- Aubinet, M., Vesala, T., & Papale, D. (2012). Eddy Covariance: A Practical Guide to Measurement and Data Analysis. Springer.
- Baldocchi, D. (2003). Assessing the eddy covariance technique for evaluating carbon dioxide exchange rates of ecosystems. *Global Change Biology*, 9, 479-492.
- Burba, G. (2013). Eddy Covariance Method for Scientific, Industrial, Agricultural, and Regulatory Applications. LI-COR Biosciences.
- Davidson, E. A., & Janssens, I. A. (2006). Temperature sensitivity of soil carbon decomposition and feedbacks to climate change. *Nature*, 440, 165-173.
- Falge, E., et al. (2001). Gap filling strategies for defensible annual sums of net ecosystem exchange. *Agricultural and Forest Meteorology*, 107, 43-69.
- Foken, T. (2008). The energy balance closure problem: an overview. *Ecological Applications*, 18, 1351-1367.
- Foken, T., & Wichura, B. (1996). Tools for quality assessment of surface-based flux measurements. *Agricultural and Forest Meteorology*, 78, 83-105.
- Kljun, N., Calanca, P., Rotach, M. W., & Schmid, H. P. (2015). A simple two-dimensional parameterisation for Flux Footprint Prediction. *Geoscientific Model Development*, 8, 3695-3713.
- Kormann, R., & Meixner, F. X. (2001). An analytical footprint model for non-neutral stratification. *Boundary-Layer Meteorology*, 99, 207-224.
- Lasslop, G., Reichstein, M., Papale, D., et al. (2010). Separation of net ecosystem exchange into assimilation and respiration using a light response curve approach. *Global Change Biology*, 16, 187-208.
- Lloyd, J., & Taylor, J. A. (1994). On the temperature dependence of soil respiration. *Functional Ecology*, 8, 315-323.
- Massman, W. J. (2000). A simple method for estimating frequency response corrections for eddy covariance systems. *Agricultural and Forest Meteorology*, 104, 185-198.
- Massman, W. J. (2001). Reply to comment by Rannik on frequency response corrections. *Agricultural and Forest Meteorology*, 107, 247-251.
- Mauder, M., & Foken, T. (2011). Documentation and instruction manual of the eddy covariance software package TK3. Universität Bayreuth.
- Moncrieff, J. B., et al. (1997). A system to measure surface fluxes of momentum, sensible heat, water vapour and carbon dioxide. *Journal of Hydrology*, 188-189, 589-611.
- Papale, D., et al. (2006). Towards a standardized processing of net ecosystem exchange measured with eddy covariance technique. *Biogeosciences*, 3, 571-583.
- Pastorello, G., et al. (2020). The FLUXNET2015 dataset and the ONEFlux processing pipeline. *Scientific Data*, 7, 225.

- Reichstein, M., et al. (2005). On the separation of net ecosystem exchange into assimilation and ecosystem respiration. *Global Change Biology*, 11, 1424-1439.
- Vickers, D., & Mahrt, L. (1997). Quality control and flux sampling problems for tower and aircraft data. *Journal of Atmospheric and Oceanic Technology*, 14, 512-526.
- Webb, E. K., Pearman, G. I., & Leuning, R. (1980). Correction of flux measurements for density effects due to heat and water vapour transfer. *Quarterly Journal of the Royal Meteorological Society*, 106, 85-100.
- Wilson, K., et al. (2002). Energy balance closure at FLUXNET sites. *Agricultural and Forest Meteorology*, 113, 223-243.
- Wutzler, T., et al. (2018). Basic and extensible post-processing of eddy covariance flux data with REddyProc. *Biogeosciences*, 15, 5015-5030.

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